

Visualizing datasets

Module 2 - Lesson 3

Overview

Visualization exposes patterns, distributions, and problems before modeling. matplotlib and seaborn build the charts that drive analysis.

Key Concepts

- Histograms reveal distributions and outliers; boxplots compare groups.
- Scatter plots expose relationships between two numeric variables.
- Bar and line charts compare categories and trends over time.
- Correlation heatmaps summarize relationships across many columns.
- Label axes, add titles, and use color deliberately; a chart must stand alone.
- Visualize before modeling and after to validate assumptions.

Example

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df["price"], bins=30)
plt.title("Course price distribution")
plt.show()

sns.scatterplot(data=df, x="duration_hours", y="enrollments")
plt.show()
```

Summary

Charts make data legible. Distributions, comparisons, and correlations guide cleaning and feature decisions - always visualize before you model.

Practice Checklist

- Plot the distribution of a numeric column and note the shape.
- Build a scatter plot of two variables and describe the relationship.
- Create a correlation heatmap for a multi-column dataset.